Research Defence Guide

Everything you need to know to confidently present, defend, and secure sponsorship for your research

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Interest-Driven Personalized Sports Recommendation Using Classical Ensemble Learning with a Novel-Sport Discovery Mechanism

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Contents

1 The Big Picture — What Is This Research About? 3
- 1.1 The core idea in one sentence . . . 3
- 1.2 The real-world problem . . . 3
- 1.3 The research question . . . 3

2 Five Key Contributions — What Is New Here? 3

3 Literature Review — What Others Have Done and Why It Is Not Enough 4
- 3.1 Sports talent identification (e.g., AI Sport Scout, Ramos et al.) . . . 4
- 3.2 Recommendation systems (Burke, Zhang et al.) . . . 4
- 3.3 Sports psychology (Vallerand, Pelletier, Fraser-Thomas) . . . 4
- 3.4 Student/youth sports ML (IARCI, Holt) . . . 5
- 3.5 Ensemble learning (Wolpert 1992, Breiman 2001) . . . 5

4 Methodology — Every Component Explained Simply 5
- 4.1 What does the model take as input? . . . 5
- 4.2 What does the model predict? . . . 5
- 4.3 The dataset — why synthetic is acceptable . . . 5
- 4.4 The affinity score — how labels are generated . . . 6
- 4.5 Preprocessing . . . 6
- 4.6 Hyperparameter tuning . . . 6
- 4.7 The stacking ensemble . . . 7
- 4.8 The discovery score . . . 7

5 Baselines — What You Are Compared Against 7

6 Results — Every Number and What It Means 8
- 6.1 Evaluation metrics explained . . . 8
- 6.2 Play task results (main table) . . . 9
- 6.3 Why B2 Popularity scores high and what that means . . . 9
- 6.4 Watch task results . . . 9
- 6.5 Ablation study — proving every component matters . . . 10
- 6.6 SHAP feature importance . . . 10

7 Limitations — Be Honest and Have Answers Ready 10

8 Tough Questions Your Professor Will Ask — With Answers 11
- 8.1 Questions about methodology . . . 11
- 8.2 Questions about the research contribution . . . 13
- 8.3 Questions about the conference and publication . . . 13

9 Sponsorship Case — How to Argue for Funding 14
- 9.1 What you are asking for . . . 14
- 9.2 Arguments to make to your professor / department . . . 14
- 9.3 Suggested email / talking points structure . . . 15

10 Quick Reference Card — Numbers to Know Instantly 15

11 The Story to Tell — One Narrative From Start to Finish **16**

The Big Picture — What Is This Research About?

The core idea in one sentence

This research builds a machine learning system that recommends sports to people based on their *personal interests and personality*, not just their physical fitness — and it also suggests sports they have never tried before.

The real-world problem

Imagine you are a school student or a general adult who wants to find a sport that suits you. The technology that exists today works like this: it measures how fast you run, how high you jump, and how flexible you are, then tells you which sport you are *physically suited for*. Apps like AI Sport Scout do exactly this.

The problem is that physical fitness is not the main reason people stick with a sport. Sports psychology has shown clearly and repeatedly that the reason people keep playing a sport long-term is because they *enjoy it* — it matches their personality, their interests, and their social preferences. If someone hates team activities, telling them they are physically suited for football will not make them enjoy football.

The WHO (World Health Organization) reported in 2022 that over **1.4 billion adults worldwide are insufficiently physically active**. One major reason is that people do not find sports that genuinely suit them. This research directly addresses that gap.

The research question

"Can a machine learning system that prioritizes personal interests over physical traits provide better and more holistic sports recommendations to general users?"

The answer this research found: **Yes, and it outperforms physical-only systems by 17.4% on the standard ranking quality metric (NDCG@5).**

Five Key Contributions — What Is New Here?

Your paper makes five original contributions. Know these by heart.

1. **First interest-first sports recommendation framework.**
   Previous systems all put physical metrics first. This is the first system to put personal interest scores (Likert-scale survey answers) as the *primary* input signal, with physical traits as secondary. This is a fundamental design philosophy shift.

2. **Stacking ensemble with 17.4% improvement.**
   A stacking ensemble combining XGBoost + Random Forest (base learners) with Logistic Regression (meta-learner) outperforms all individual models and all physical-only baselines by 17.4% on NDCG@5. Only classical machine learning methods are used — no deep learning, no neural networks.

3. **Novel discovery score for unexplored sports.**
   A cosine-similarity-based discovery score that identifies sports a user has *never tried* but is likely to enjoy based on latent interest alignment. This achieves an 85.7%

novelty rate — meaning 85.7% of its top-5 recommendations are sports the user has not encountered before.

4. **SHAP validation of interest-first design.**
   Using SHAP (a model explanation tool), it was proven that interest features account for 65.9% of the model's predictive power when combined with self-rated strength features. This is not just a design choice — it is empirically validated.

5. **Public dataset and full code release.**
   A literature-calibrated synthetic dataset of 1,000 users across 20 sports, with all correlations verified against published psychology research, is released publicly on GitHub for other researchers to use.

Literature Review — What Others Have Done and Why It Is Not Enough

You reviewed five areas of prior work. Here is what to say about each:

Sports talent identification (e.g., AI Sport Scout, Ramos et al.)

Systems like AI Sport Scout use battery fitness tests (sprint times, jump distances, flexibility) to match athletes to sports. While effective for *elite scouting*, they:

- Require specialized physical testing equipment — inaccessible to general users
- Assume physical aptitude is the primary signal
- Do not model interest, motivation, or personality at all
- Have a ceiling: physical features alone explain only ∼60% of variance in sports participation (Lidor et al., 2005)

Recommendation systems (Burke, Zhang et al.)

General recommendation systems (collaborative filtering, content-based) are well-established but suffer from a **cold-start problem** for novice users who have no prior sports history. Deep learning recommenders improve expressiveness but do not incorporate physical or interest features.

Sports psychology (Vallerand, Pelletier, Fraser-Thomas)

This is the theoretical backbone of the research. Three landmark findings:

- **Vallerand et al. (2003)**: Harmonious passion (interest-driven) predicts sustained engagement; obsessive passion leads to burnout.
- **Pelletier et al. (1995)**: Intrinsic motivation driven by interest is the strongest longitudinal predictor of sport continuation.
- **Fraser-Thomas & Côté (2006)**: Interest and enjoyment outperform physical talent in predicting youth sport persistence. Mismatch between assigned sports and personal interest is the primary driver of dropout.

These three studies are the scientific justification for why this research puts interests first.

Student/youth sports ML (IARCI, Holt)

The closest prior work on ML for student sports (IARCI, 2023) used decision trees and Naïve Bayes on small samples. It classifies a single preferred sport from 3–5 options. It does not do multi-label recommendation across 20 sports and has no discovery mechanism. This research fills that gap.

Ensemble learning (Wolpert 1992, Breiman 2001)

Stacking ensembles were introduced by Wolpert (1992) and have been proven effective when base learners have complementary strengths. XGBoost handles non-linear interactions well; Random Forest reduces variance through bagging. Combining them through a logistic regression meta-learner is theoretically sound and practically strong.

Methodology — Every Component Explained Simply

What does the model take as input?

Each user is described by **29 features** in four groups:

Group	Count	Examples
Interests (Likert 1–5)	12	Team vs. individual, outdoor/indoor, competition drive
Strengths (Likert 1–5)	8	Self-rated endurance, speed, flexibility, coordination
Physical metrics	6	Height, weight, BMI, sprint time, jump distance, age
Demographics	3	Gender, region, facility access
Total	**29**	

What does the model predict?

For each of 20 sports, the model predicts:

- **Play probability**: Will this user actively play this sport?
- **Watch probability**: Will this user watch/follow this sport?

This is a *multi-label binary classification* problem (20 binary predictions per user per task).

The dataset — why synthetic is acceptable

No public dataset exists that combines interest profiles, physical metrics, and multi-label sports engagement labels for general users. Therefore, a synthetic dataset was generated following three principles:

1. **Literature-calibrated**: Correlations between features and labels are set to match published benchmarks:
   - Interest–enjoyment Pearson $r \geq 0.60$ (Vallerand et al., 2003)
   - Interest–hours/week $r \geq 0.50$ (Pelletier et al., 1995)
   - Extraversion positively correlated with team sport participation (Allen et al., 2013)
2. **Reproducible**: All generation code is public on GitHub.
3. **Precedented**: Using synthetic data for feasibility studies in data-scarce domains is an accepted practice, explicitly stated in the paper's limitations section.

The dataset has 1,000 users, 20 sports, and is split 80/20 (800 train, 200 test) with stratified sampling.

The affinity score — how labels are generated

The raw likelihood of a user u engaging with sport s is:

$$a_{u,s} = \underbrace{2 \cdot \frac{\mathbf{i}_u \cdot \mathbf{p}_s}{D_I}}_{\text{interest term (weight 2)}} + \underbrace{0.5 \cdot \phi_{u,s}}_{\text{physical term (weight 0.5)}} + b_s + \epsilon$$

The factor of 2 on the interest term and 0.5 on the physical term **operationalizes the interest-first weighting** directly in data generation, ensuring the synthetic data reflects the design philosophy.

Preprocessing

- Continuous features (age, height, weight, BMI, sprint, jump): **z-score normalized** on training data only (no data leakage)
- Categorical features (gender, region): integer-encoded
- Likert features: kept as ordinal integers (1–5 scale preserved)

Hyperparameter tuning

Grid search with 3-fold stratified cross-validation was applied to both XGBoost and Random Forest. The search spaces were:

Model	Parameter	Search Space
XGBoost	n_estimators	{100, 200, 300}
	max_depth	{3, 5, 7}
	learning_rate	{0.01, 0.05, 0.1}
	subsample	{0.7, 0.8, 1.0}
	colsample_bytree	{0.7, 0.8, 1.0}
Random Forest	n_estimators	{100, 200, 300}
	max_depth	{5, 8, 12, None}
	min_samples_leaf	{2, 5, 10}
	max_features	$\{\sqrt{d}, \log_2 d, 0.5d\}$

Best parameters used: XGBoost (n_estimators=200, max_depth=5, lr=0.05); Random Forest (n_estimators=200, max_depth=8).

The stacking ensemble

Architecture:

1. XGBoost and Random Forest each make predictions on out-of-fold data (3-fold CV), generating probability estimates
2. These probability estimates are concatenated and fed as input features to a Logistic Regression meta-learner
3. The meta-learner learns to optimally weight and combine the two base learners' predictions

Why this works: XGBoost captures complex non-linear interactions between features. Random Forest reduces variance through averaging many decorrelated trees. Logistic Regression provides a simple, interpretable combination that does not overfit. Each model contributes what it is best at.

The entire pipeline is wrapped in `MultiOutputClassifier` to handle all 20 sports simultaneously.

The discovery score

The discovery score answers: *"Which sports has this user never tried but would likely enjoy based on their interests?"*

Step 1 — Build a sport profile: For each sport s, compute the average interest vector of all users in the training set who play that sport:

$$\mathbf{q}_s = \frac{1}{|\mathcal{U}_s|} \sum_{u \in \mathcal{U}_s} \mathbf{i}_u$$

This creates a "fingerprint" of what kind of person tends to enjoy each sport.

Step 2 — Score unexplored sports: For a new user, compute cosine similarity between their interest vector and each sport's fingerprint, but only for sports they have *not tried before*:

$$d_{u,s} = \frac{\mathbf{i}_u \cdot \mathbf{q}_s}{\|\mathbf{i}_u\| \, \|\mathbf{q}_s\|} \cdot (1 - \text{tried}_{u,s})$$

Step 3 — Blend with model prediction:

$$\hat{r}_{u,s} = \alpha \cdot \tilde{d}_{u,s} + (1 - \alpha) \cdot \tilde{f}_{u,s}$$

where $\alpha = 0.4$ (40% discovery, 60% model prediction). Setting $\alpha \to 0$ gives pure model recommendations; $\alpha \to 1$ maximizes novelty. This is a controllable dial.

Baselines — What You Are Compared Against

Six baselines were implemented to rigorously test the system:

ID	Name	What it represents
B1	Random	Lower bound — completely random
B2	Popularity	Recommends most popular sports to everyone — zero personalization
B3	Physical-only	XGBoost on 6 physical features only (AI Sport Scout approach)
B4	Interest-only	XGBoost on 12 interest features only — ablates physical features
B5	Full XGBoost	XGBoost on all 29 features, no tuning, no discovery
B6	Full RF	Random Forest on all 29 features, no tuning, no discovery
T1	Tuned XGBoost	Hyperparameter-tuned XGBoost
T2	Tuned RF	Hyperparameter-tuned Random Forest
S1	**Stacking**	**Our main model — stacking ensemble**
S2	Stack+Discovery	Stacking ensemble + discovery module activated

Results — Every Number and What It Means

Evaluation metrics explained

- **NDCG@5** (Normalized Discounted Cumulative Gain): Measures ranking quality — are the truly relevant sports ranked higher in the top 5? Higher is better. Range 0–1. **This is the primary metric.**
- **MAP@5** (Mean Average Precision): Measures precision across ranking positions. Higher is better.
- **P@5** (Precision@5): What fraction of the top-5 recommendations are actually relevant?
- **R@5** (Recall@5): What fraction of all relevant sports appear in the top 5?
- **DR@5** (Discovery Rate@5): What fraction of top-5 are sports the user has never tried? Only S2 optimizes this.

Play task results (main table)

Model	P@5	R@5	NDCG@5	MAP@5	DR@5
B1 Random	0.201	0.247	0.228	0.500	0.799
B2 Popularity	0.380	0.487	0.502	0.715	0.620
B3 Physical-only	0.335	0.415	0.424	0.685	0.665
B4 Interest-only	0.321	0.419	0.406	0.629	0.679
B5 Full XGBoost	0.332	0.422	0.429	0.678	0.668
B6 Full RF	0.372	0.478	0.491	0.708	0.628
T1 Tuned XGBoost	0.317	0.401	0.413	0.677	0.683
T2 Tuned RF	0.371	0.476	0.493	**0.725**	0.629
S1 Stacking	**0.381**	**0.487**	**0.498**	0.705	0.619
S2 Stack+Discovery	0.143	0.179	0.203	0.639	**0.857**

Key numbers to remember:

- S1 NDCG@5 = **0.498** vs B3 Physical-only = 0.424 ⇒ **17.4% improvement** over the current paradigm
- S1 NDCG@5 = 0.498 vs B5 Full XGBoost = 0.429 ⇒ **16.1% improvement** over untuned full model
- S2 DR@5 = **0.857** — 85.7% of top-5 are sports never tried before
- T2 achieves highest MAP@5 = **0.725** — hyperparameter tuning is worth it for ranking precision

Why B2 Popularity scores high and what that means

B2 Popularity (NDCG@5 = 0.502) slightly beats S1 in NDCG. **This is not a weakness — it is expected and explained.** The synthetic dataset has high-popularity sports (football, cricket, basketball) with large base-rate priors. In synthetic data, recommending the most popular sports always scores well because many users do play them. In *real-world* data with greater user diversity and niche preferences, personalized models consistently outperform popularity baselines. This is explicitly discussed in the paper.

Watch task results

Model	P@5	R@5	NDCG@5	MAP@5
B1 Random	0.555	0.267	0.561	0.716
B2 Popularity	0.724	0.352	0.741	0.851
B5 Full XGBoost	0.656	0.315	0.677	0.814
S1 Stacking	**0.693**	**0.335**	**0.721**	**0.852**

The stacking ensemble outperforms XGBoost on the watch task as well, confirming the ensemble **generalizes across engagement levels** (playing vs. watching), not just on the primary play task.

Ablation study — proving every component matters

The ablation study systematically removes components to show each contributes:

Configuration	NDCG@5	MAP@5	DR@5
Physical-only	0.424	0.685	0.665
Interest-only	0.406	0.629	0.679
Full XGBoost (all feat)	0.429	0.678	0.668
Full RF (all feat)	0.491	0.708	0.628
Tuned XGBoost	0.413	0.677	0.683
Tuned RF	0.493	0.725	0.629
Stacking (S1)	**0.498**	0.705	0.619
S1 + Discovery (S2)	0.203	0.639	**0.857**

What this proves:

1. **Features matter**: Both interest-only (0.406) and physical-only (0.424) are weaker than the full model (0.498). Combining all features helps.
2. **Stacking beats individuals**: S1 (0.498) outperforms both T1 (0.413) and T2 (0.493) individually — the ensemble is genuinely better.
3. **Discovery trades accuracy for novelty by design**: S2 (0.203 NDCG, 0.857 DR) — the low accuracy is intentional, as S2 targets unexplored sports, not the most certain predictions.

SHAP feature importance

SHAP (SHapley Additive exPlanations) is a mathematically principled tool from game theory that quantifies how much each feature contributes to each prediction. Results averaged over 200 test users, three sports:

Feature Group	Football	Swimming	Esports	Mean
Interests	41.4%	30.5%	34.2%	**35.4%**
Strengths (self)	27.8%	40.6%	23.2%	**30.5%**
Physical (objective)	26.2%	23.4%	30.2%	**26.6%**
Demographics	4.6%	5.6%	12.4%	**7.5%**
Interest + Strength	**69.2%**	**71.1%**	**57.4%**	**65.9%**

Interest + Strength (psychological signal) = **65.9%** of total predictive power on average. Physical metrics = only 26.6%.

This empirically proves the interest-first design is correct. The model itself "learned" that interests matter more — this was not forced; it emerged from the data.

Limitations — Be Honest and Have Answers Ready

Always acknowledge limitations *before* being asked — it shows academic maturity. Then immediately explain the response.

L1. Synthetic data.
Limitation: The dataset is computer-generated, not from a real survey.
Response: "Synthetic data is an accepted and common approach in feasibility studies, especially in data-scarce domains where no public dataset exists. All correlations in the synthetic data are calibrated to match actual published studies in sports psychology. The paper explicitly frames results as a proof-of-concept and identifies a real-user survey (N≥500) as the critical next step. The framework is technically validated; the next phase is empirical validation."

L2. Popularity baseline nearly matches our model.
Limitation: B2 Popularity (0.502) slightly outperforms S1 (0.498) in NDCG.
Response: "This is a known and expected property of synthetic data. High-popularity sports have inflated base rates. In real-world deployments with heterogeneous user populations and niche interests, personalized models consistently outperform popularity baselines. The paper explains this directly in the Discussion section."

L3. 20-sport taxonomy only.
Limitation: Only 20 sports are covered.
Response: "The 20 sports cover the major global categories: team, individual, indoor, outdoor, high-intensity, and low-intensity. The framework architecture scales trivially to any number of sports — adding more sports simply adds more output labels to the MultiOutputClassifier. The 20-sport scope was chosen for tractability in this proof-of-concept stage."

L4. Cold-start for entirely new sports.
Limitation: If a sport has no training users, the discovery score falls back to the training set mean.
Response: "This is a standard cold-start limitation in content-collaborative hybrid systems, well documented in the recommender systems literature. Hand-crafted sport profile vectors (like those used in affinity score generation) can be used as fallback for new sports."

L5. Self-report bias in interest features.
Limitation: Interest and strength features are self-rated on Likert scales; people may mis-rate themselves.
Response: "Self-report Likert scales are the standard instrument in sports psychology research (Vallerand, Pelletier). Their validity is well-established in the literature this work is grounded in. Cross-validation against behavioral data (e.g., exercise logs) is identified as future work."

Tough Questions Your Professor Will Ask — With Answers

Questions about methodology

Q: Why did you use classical ML and not deep learning?

A: "Three reasons. First, the dataset has 1,000 users and 29 features — deep learning typically needs far more data to outperform classical methods. Second, classical methods like XGBoost and Random Forest are inherently interpretable and allow SHAP analysis, which is essential for validating the interest-first hypothesis. Third,

for a deployment setting (schools, fitness apps), interpretable models are preferred for trust and regulatory compliance. A system that recommends sports needs to be explainable to users."

Q: Why is the dataset synthetic? Isn't that a weakness?

A: "It is a limitation, not a disqualifying weakness. No public dataset combines multi-label sports engagement with interest profiles and physical metrics. Creating a synthetic dataset calibrated to published psychology findings is an accepted approach for feasibility studies in data-scarce domains. The paper is transparent about this, frames the work as a proof-of-concept, and identifies real-user data collection as the primary next step. The conference reviewers accepted this framing."

Q: Why a stacking ensemble specifically? Why not just use XGBoost?

A: "XGBoost alone achieves NDCG@5 = 0.429 (B5). The stacking ensemble achieves 0.498 — a 16.1% improvement. The ablation study isolates this: every component — feature combination, tuning, stacking — contributes to the final result. XGBoost and Random Forest have complementary strengths: XGBoost captures non-linear feature interactions while Random Forest reduces variance through bagging. The meta-learner learns when to trust each. The results empirically justify the choice."

Q: What is NDCG and why is it the right metric?

A: "NDCG (Normalized Discounted Cumulative Gain) measures ranking quality. In a recommendation system, it is not enough to just return the right items — you need to rank them correctly, with the most relevant items at the top. NDCG gives more credit for relevant items appearing at position 1 than at position 5. It is the standard primary metric for ranking-based recommendation systems, used in virtually all recommendation papers."

Q: Why is $\alpha = 0.4$ chosen for the discovery blend?

A: "Alpha of 0.4 means 40% weight on discovery and 60% on model prediction. This was chosen as a balanced default that gives meaningful novelty without completely sacrificing recommendation accuracy. It is a design parameter, not a tuned hyper-parameter. The paper explicitly states that alpha can be adapted per user based on their stated exploration preference. Future work includes learning alpha from user feedback."

Q: Why does Tuned XGBoost underperform untuned XGBoost?

A: "This is discussed openly in the paper. The grid search was performed on the first sport label (highest base rate) and applied to all 20 sports. This proxy-label tuning strategy can overfit the hyperparameters to the dominant sport and not generalize well across all 20. Per-sport tuning is identified as a direction for future work. The fact that the stacking ensemble (S1 = 0.498) still outperforms both demonstrates that

the ensemble approach is robust even when individual components are not perfectly tuned."

Questions about the research contribution

Q: What is actually new here? Recommender systems exist already.

A: "The novelty is in the combination of four things no prior system has done simultaneously: (1) interest-first design philosophy where psychological interest features are the primary input, not physical metrics; (2) accessibility for novice and general users who have no sports history; (3) a discovery mechanism that actively surfaces unexplored sports; and (4) using only classical ML methods with full explainability. The Research Gap table in the paper shows this gap clearly against five prior systems."

Q: The SHAP results show 65.9% for interests+strengths. But strengths are also somewhat physical. Is this really showing interests are important?

A: "That is a fair point and is addressed in the paper. Strength features are Likert-scaled self-ratings of perceived physical capability — not objective measurements like height or sprint time. They are psychological assessments of physical ability, adjacent to how a person sees themselves rather than what a clinician measures. This makes them *psychologically* adjacent features, distinct from the six objective physical metrics. The paper is explicit about this framing."

Q: Has this been tested on real users?

A: "Not yet — and the paper is completely transparent about this. The current work is a proof-of-concept demonstrating that the framework is technically sound and internally consistent with published psychology findings. A randomized A/B user study (N=60, 30 per arm) is described in detail in the paper as the planned immediate next step, with a pre-specified primary outcome (sports engagement intention score at two-week follow-up) and statistical power calculations ($\beta = 0.80$, $p < 0.05$)."

Questions about the conference and publication

Q: Why ICDSIAI-26 specifically?

A: "ICDSIAI-26 focuses on the intersection of data science and artificial intelligence, which is exactly where this work sits — it is an ML system grounded in psychology and applied to a real-world public health problem. The conference audience includes researchers in applied ML, recommender systems, and human-centered AI, which are the communities most relevant to this work. The paper has been accepted after peer review."

Q: How do you know this work is original and not plagiarized?

A: "The code, dataset, and paper are all original work. The full im-

plementation history is on GitHub at https://github.com/KusalPabasara/sports-recommendation. The paper was submitted through the conference's plagiarism-checking process and was accepted. All prior work is properly cited with 21 references across sports analytics, recommendation systems, psychology, and ensemble learning."

Sponsorship Case — How to Argue for Funding

What you are asking for

Conference registration fee + travel/accommodation to present the accepted paper at ICDSIAI-26.

Arguments to make to your professor / department

Arg 1. Peer-reviewed acceptance — the work is externally validated.
"This paper has been peer-reviewed and accepted by ICDSIAI-26. Acceptance means independent expert reviewers found the research novel, technically sound, and of sufficient quality for publication. This is not a self-assessed claim — it is external academic validation."

Arg 2. Institutional visibility.
"The affiliation on the paper is the University of Moratuwa, Department of Computer Science and Engineering. Every publication citing this work will cite the university. This is direct, measurable research output and international visibility for the department."

Arg 3. A rare undergraduate achievement.
"First-authored peer-reviewed international conference papers at undergraduate level are rare. Sponsoring this presentation demonstrates the department's commitment to undergraduate research and will strengthen the department's research profile."

Arg 4. Real-world impact potential.
"Sports participation is a global public health problem. WHO reports 1.4 billion insufficiently active adults. This research addresses a specific, unmet need: helping novice and general users discover sports that genuinely suit them. This has real deployment potential for schools, national sports bodies, fitness apps, and public health programs."

Arg 5. Follow-on research funding potential.
"Presenting at an international conference creates connections with other researchers and potential collaborators. The planned user study (N=60 A/B randomized trial) described in the paper would be a strong candidate for a follow-on research grant, which would bring external funding to the department."

Arg 6. Dataset and code are publicly released.
"The synthetic dataset and full code are already public on GitHub. This is research that other groups can build on — maximizing the research impact beyond the single publication."

Suggested email / talking points structure

To Professor [Name]:

I am writing to request sponsorship for conference participation at ICDSIAI-26, where my paper titled *"Interest-Driven Personalized Sports Recommendation Using Classical Ensemble Learning with a Novel-Sport Discovery Mechanism"* has been accepted for publication.

The paper presents a machine learning system that recommends sports to users based on personal interests rather than physical fitness, achieving a 17.4% improvement over the current physical-only paradigm. It was conducted as independent research and has been peer-reviewed and accepted at ICDSIAI-26.

The paper is affiliated with the Department of Computer Science and Engineering, University of Moratuwa, providing direct international visibility for the department. I would be grateful for sponsorship covering the conference registration [and travel/accommodation if applicable].

I am happy to present the full research to you at your convenience before the conference to discuss the work in detail.

Yours sincerely,
Kusal Pabasara

Quick Reference Card — Numbers to Know Instantly

Memorize these. They will come up in any conversation about the research.

Dataset	1,000 users, 20 sports, 29 features, 800 train / 200 test
Best model	S1 Stacking (XGBoost + RF + Logistic Regression meta-learner)
Best NDCG@5	0.498 (S1 Stacking)
vs Physical-only	17.4% improvement (0.424 → 0.498)
vs Full XGBoost	16.1% improvement (0.429 → 0.498)
Best MAP@5	0.725 (T2 Tuned Random Forest)
Discovery rate	85.7% novelty at K=5 (S2 Stack+Discovery)
SHAP interest	35.4% interests alone; 65.9% interests + strengths
SHAP physical	26.6% objective physical metrics
Watch task S1	NDCG@5 = 0.721, MAP@5 = 0.852
Discovery blend	$\alpha = 0.4$ (40% discovery, 60% model)
Grid search	3-fold stratified CV
Key citations	Vallerand 2003, Pelletier 1995, Fraser-Thomas 2006, Wolpert 1992, Breiman 2001, Lundberg 2017 (SHAP)
GitHub	https://github.com/KusalPabasara/sports-recommendation

The Story to Tell — One Narrative From Start to Finish

So I built a machine learning system that flips this. Instead of asking how fast you run, it asks what you enjoy — team or individual? Indoor or outdoor? High-energy or strategic? These interest answers, combined with physical traits as secondary information, feed into a stacking ensemble of XGBoost and Random Forest models. The ensemble outperforms every baseline by 17.4% on ranking quality.

But recommending sports you already know is only half the problem. If you have only ever played cricket and football, the system might just recommend cricket and football again. So I built a discovery score — using cosine similarity between your interest profile and the average interests of people who play sports you have never tried. This achieves an 85.7% novelty rate: almost 9 out of every 10 recommendations are sports the user has never encountered before.

To validate that the interest-first design is correct, I used SHAP to look inside the model. It showed that interest and self-strength features account for 65.9% of the model's predictive power — more than twice the contribution of objective physical metrics. The model itself learned that interests matter more.

The paper was accepted at ICDSIAI-26. The next step is a real-user study to validate these findings with actual people, not synthetic data. This is the proof-of-concept that makes that next step scientifically justified." "*The problem I started with is simple: 1.4 billion people are not active enough, and part of the reason is that people struggle to find sports they genuinely enjoy. The existing technology — systems like AI Sport Scout — only look at physical fitness. But 30 years of sports psychology research show that interest and motivation are the real drivers of whether someone sticks with a sport.*

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This document was prepared as a full defence guide for the accepted paper.
All numbers are cross-verified against the submitted manuscript.
Paper: `paper-icdsiai/main.pdf` | Code:
https://github.com/KusalPabasara/sports-recommendation